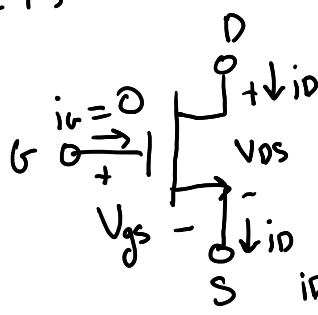
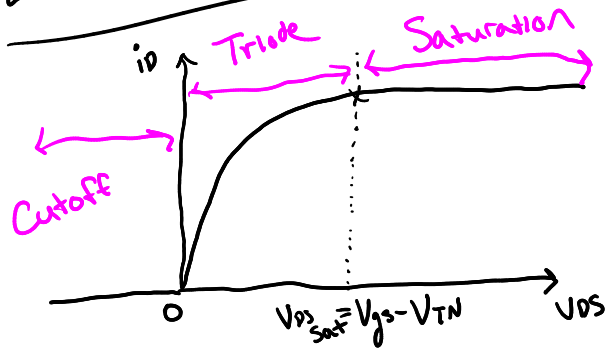


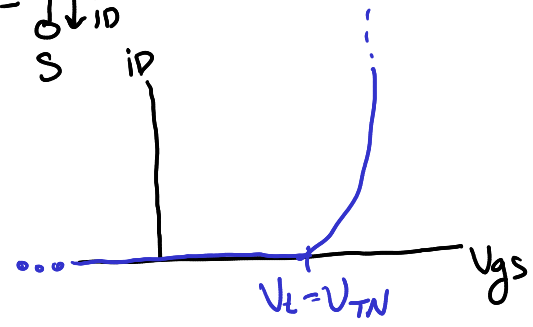
Lecture 17

MOSFETs

* + 1 point to all for Exam 2



D - Drain
G - Gate
S - Source



Cutoff $V_{GS} < V_{TN}$
 $i_D = 0$ regardless of V_{DS}

Triode $V_{GS} > V_{TN}$ and $V_{DS} < (V_{GS} - V_{TN})$

$$i_D = \underbrace{K_n}_{\text{Constant}} \underbrace{\frac{W}{L}}_{\text{Capitol}} \left[(V_{GS} - V_{TN})V_{DS} - \frac{1}{2}V_{DS}^2 \right]$$

$$V_{OV} \text{ Overdrive}$$

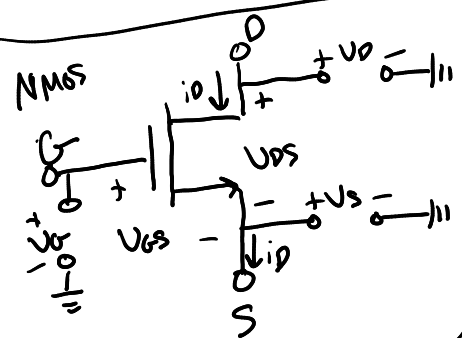
Saturation $V_{GS} > V_{TN}$ and $V_{DS} > (V_{GS} - V_{TN})$

$$i_D = \frac{1}{2} K_n \frac{W}{L} \left[(V_{GS} - V_{TN})^2 \right] (1 + \lambda V_{DS})$$

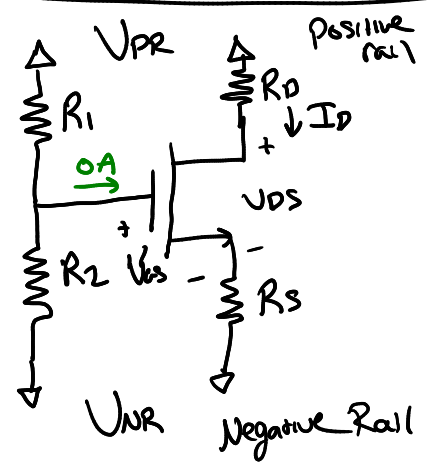
We will not consider this

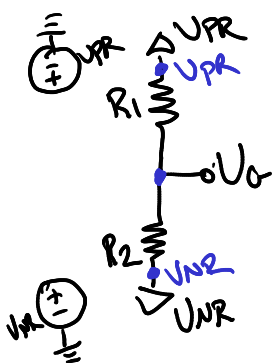
PMOS

$K_p = \frac{1}{2} k_p' \frac{W}{L}$



NMOS @ DC
 $V_{DS} \rightarrow V_{DS}, i_D \rightarrow I_D$
 $V_{GS} \rightarrow V_{GS}$
 "Gate" doesn't depend on Cutoff, triode, or Sat.
Operating mode





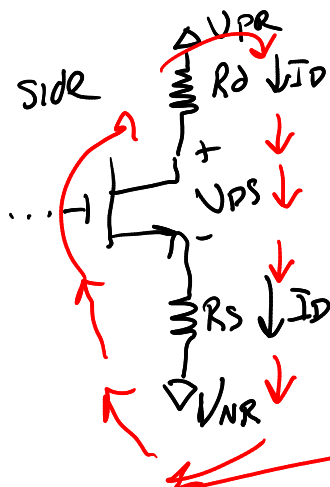
KCL @ V_G

$$\frac{1}{R_1}(V_G - V_{PR}) + \frac{1}{R_2}(V_G - V_{NR}) = 0$$

$$\Rightarrow V_G = \frac{R_2 V_{PR} + R_1 V_{NR}}{R_1 + R_2} \dots$$

V_G doesn't depend on "mode"

Next, on DS side

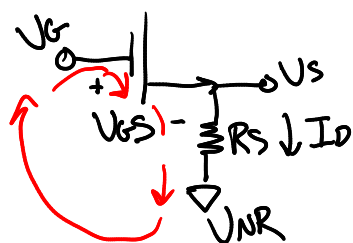


KVL on DS side

$$-V_{PR} + R_D I_D + V_{PS} + R_S I_D + V_{NR} = 0$$

does not depend on cut off, triode, sat.

GS KVL



$$-V_G + V_{GS} + R_S I_D + V_{NR} = 0$$

also

$$V_{GS} = V_G - V_S$$

Assume SAT., $V_{GS} > V_{TN}$ and $V_{DS} > V_{ov} \Rightarrow V_{DS} > (V_{GS} - V_{TN})$

$$I_D = K_N (V_{GS} - V_{TN})^2$$

Substitute into GS KVL

in analysis... looking for I_D, V_{GS}, V_{DS}

$$-V_G + V_{GS} + R_S [K_N (V_{GS} - V_{TN})^2] + V_{NR} = 0$$

V_{GS} is only unknown

$$aV_{GS}^2 + bV_{GS} + c = 0$$

Suppose ... $V_{PR} = +15V, V_{NR} = -15V, R_1 = 900k\Omega, R_2 = 100k\Omega$
 $V_G = -12V, K_N = 2 \frac{mA}{V^2} = 2 \times 10^{-3} \frac{A}{V^2}, V_{TN} = 0.5V$
 $R_D = 25k\Omega, R_S = 4k\Omega$

$$V_{GS} = \underline{\underline{V_{GS1} = +1V}}, \quad V_{GS2} = -0.125V$$

Also must have, $V_{DS} > V_{GS} - V_{TN} = 1 - 0.5$
 $V_{DS} > 0.5V$

Use KVL for DS loop

$$-V_{PR} + R_D I_D + V_{DS} + R_S I_D + V_{NR} = 0$$

Q3
Q4 grade

$$\text{and } \checkmark I_D = K_N (\checkmark V_{GS} - \checkmark V_{TN})^2 \Rightarrow I_D = 0.5 \text{ mA} \checkmark$$

$$-15 + (25 \times 10^3)(0.5 \times 10^{-3}) + V_{DS} + (4 \times 10^3)(0.5 \times 10^{-3}) - 15 = 0$$

$$\Rightarrow V_{DS} = 15.5 \text{ V}$$

$$\text{is } V_{DS} > V_{GS} - V_{TN} \\ \text{yes } 15.5 > 0.5$$